ACP

AARAMBH CHAPTERWISE PROBLEMS

Class-9th

TISSUES

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TISSUES

(Options acche se padhna)

MULTIPLE CHOICE QUESTIONS (1 Mark)

1. Which of the following is not a type of plant tissue?
A) Epithelial tissue
B) Meristematic tissue
C) Permanent tissue
D) Vascular tissue

2. The tissue responsible for providing support and mechanical strength to plants is:
A) Epithelial tissue
B) Meristematic tissue
C) Connective tissue
D) Parenchyma tissue

3. A permanent tissue of collenchyma provides A....B....
A) food , water
B) flexibility , mechanical support
C) support , water
D) water , buoyancy

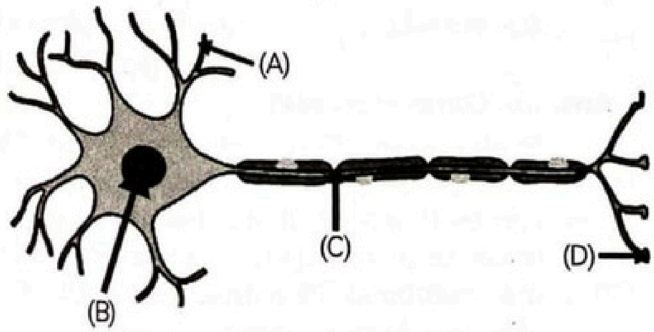
4. Girth of stem increases due to:
(a) Apical meristem
(b) lateral meristem
(c) intercalary meristem
(d) vertical meristem

5. The tissue responsible for the movement of the body and its parts is:
A) Epithelial tissue
B) Nervous tissue
C) Muscular tissue
D) Connective tissue

6. A nail is inserted into the trunk of a tree at a height of 1 metre from the ground level. After 3 years, the nail will:
(a) move downwards
(b) move upwards
(c) remain at the same position
(d) move sideways

7. Select the row which shows the correct labelling for parts.

	A	B	C	D
(a)	Nucleus	Dendrite	Axon	Nerve ending
(b)	Dendrite	Nucleus	Axon	Nerve ending
(c)	Nucleus	Dendrite	Nerve ending	Axon
(d)	Dendrite	Cell body	Axon	Nerve ending



8. Xylem and phloem are types of:
A) Epithelial tissue
B) Nervous tissue
C) Muscular tissue
D) Plant tissues

9. Light and dark bands are visible in:
A) cardiac muscle
B) smooth muscles
C) striated muscles
D) unstriated muscles

10. . Which statement below is incorrect?
A) Some plant tissues continue to divide throughout their lifespan.
B) Animals generally have fewer dead tissues compared to plants.
C) Cells in animals tend to be more uniform compared to plants.
D) There is clear demarcation between dividing and non-dividing regions in animals.

11. in desert plant ,rate of water loss get reduced due to presence of
A) Cutin
B) lignin
C) Suberin
D) stomata

12. Which connective tissue connects muscles to bones?
A) Ligaments
B) Cartilage
C) Tendons
D) Areolar

13. What is the main function of meristematic tissue?
(a) Growth
(b) Photosynthesis
(c) Respiration
(d) Transpiration

14. If the intestine absorbs the digested food materials, which type of epithelial cells are responsible?
a) Spindle fibres
b) Cuboidal epithelium
c) Stratified squamous epithelium
d) Columnar epithelium

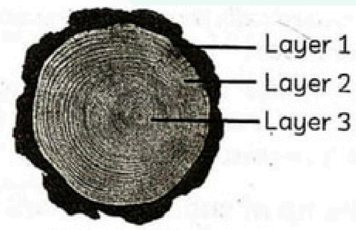
15. While doing work and running, you can move your organs like hands, legs, etc. So which among the following is correct?
a) Smooth muscles can contract and pull the ligament to move the bones.
b) Smooth muscles can contract and helps to pull the tendons.
c) Skeletal muscles contract and helps to pull the ligament to move the bones.
d) Skeletal muscles contract and help to pull the tendon to move the bones.

16. Bone matrix is rich in:
a) calcium and potassium
b) phosphorus and potassium
c) fluoride and calcium
d) calcium and phosphorus

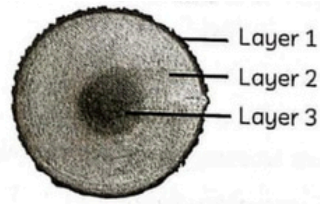
17. Fats are stored in which tissue of the human body:
a) bones
b) cartilage
c) cuboidal epithelium
d) adipose tissue

18. Cork cells are made from waterproof water and gases by the presence of:
a) cellulose
b) lipids
c) Suberin
d) lignin

19. The picture shows the cross-sections of two chopped tree trunks.



Cross-section of tree trunk (I)



Cross-section of tree trunk (II)

Which layers in both the cross-sections of chopped tree trunks are made up of dead cells? Select the correct option.
Cross-section of tree (I) - Cross-section of tree (II)
(a) Layer 1 Layer 1
(b) Layer 1 Layer 2
(c) Layer 3 Layer 3
(d) Layer 3 Layer 1

20. Blood is a connective tissue which has a fluid (liquid) matrix called plasma in which RBCs, WBCs and platelets are suspended. The plasma contains proteins, salts and hormones. Blood flows and transports gases, nutrients, hormones and waste materials to different parts of the body. Haemoglobin is present in which component of the blood:
(a) RBCs
(b) WBCs
(c) platelets RBCs
(d) plasma

21. What is found in the fluid part of blood after the removal of corpuscles?
(a) Plasma
(b) Lymph
(c) Platelets
(d) WBC

(pahle points socho firr likho)

SHORT ANSWER TYPE QUESTIONS (2 - 3 Marks)

22. Differentiate sclerenchyma and parenchyma tissues.

23. Give reasons for the following statements:
(a) Meristematic cells lack vacuole, despite having a prominent nucleus.
(b) In sclerenchymatous tissues, intercellular spaces are absent.
(c) Pear fruit gives us a crunchy, granular sensation when we chew it.
(d) Tree branches can move freely and bend in high winds.
(e) Removing the coconut tree's husk can be challenging.

24. Name four functions of areolar tissue.

25. Why blood is considered as a connective tissue despite being a fluid?

26. What are skeletal connective tissues, and mention three functions?

27. Explain, in short, the key difference between simple and complex issues:

28. A plant has flexible branches and rough fruit texture. Identify the tissues responsible and explain their roles briefly.

29. Explain the structure, function and location of nervous tissue.

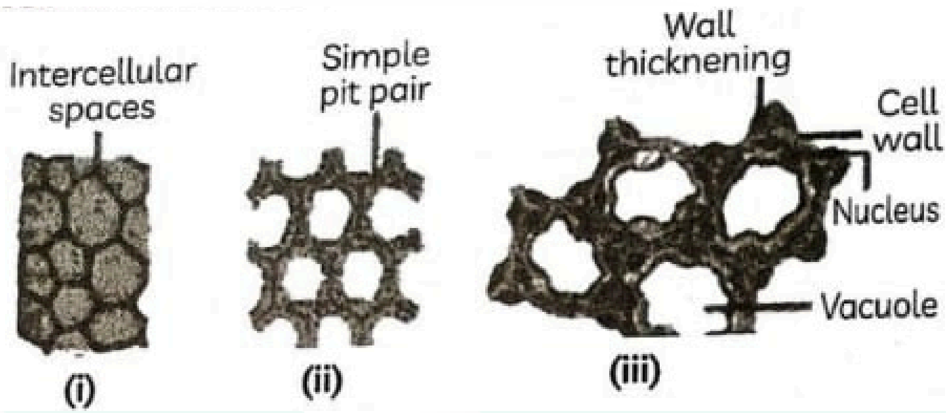
30. with the help of diagram show the difference between straited muscle , skeletal muscle and cardiac muscle .

31. Name the diffrent type of meristematic tissue and draw and show their location

32. During a visit to the lakeside, Ritik saw a water hyacinth floating on the surface of the water. He was surprised to see plants floating without roots anchored to the soil and wondered why this happened. What could be the reason for these plants floating on the water?

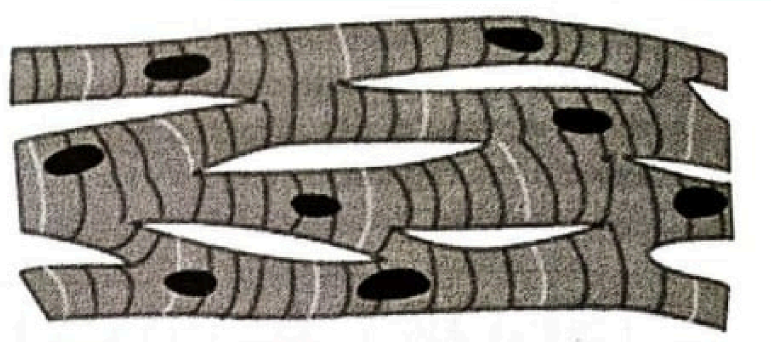
33.Give reasons for:
(A) We get a crunchy and granular feeling when we chew pear fruit.
(B) Branches of a tree move and bend freely in high wind velocity.
(C) Intercellular spaces are absent in sclerenchymatous tissues.

. 34. Given below are three figures (i), (ii) and (iii).



(A) Identify figures (i), (ii) and (iii).
(B) Which one is commercially used to get jute and hemp?
(C) Which one is modified to store products?
(D) Which one of these provide both mechanical strength and flexibility?

35. The image shows the microscopic view of a type of muscle.



(A) Write two structural features of this tissue.
(B) State the location of this tissue in the human body.
(C) What will happen if these muscles undergo fatigue?

36. Belonging to a rural family background, Jaspreet while enjoying coconut water with her friend Seema, told her that it is difficult to pull out the husk of a coconut tree to which Seema replied that it is due to the presence of sclerenchyma Can you support the statement made by Seema by providing a suitable reason?

(pahle points socho firr likho)

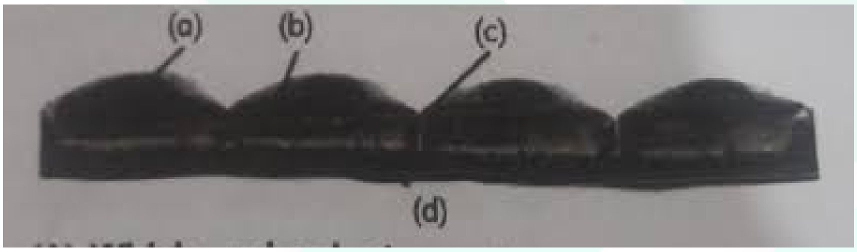
Long Answer Type Questions (5 Marks)

37. Show the types of animal tissue using flow chart .

38. what is connective tissue ? explain its types .

39. What is a permanent tissue? Classify permanent tissue and describe them.

40. A teacher shows a type of animal tissue in two different shapes and asks the following questions related to it to the students in class.
(A) Which animal tissue is represented in the figure given above?
(B) Label figure (i) to (iv).
(C) Describe the tissue and its function.
(D) What are the different types of animal tissues?



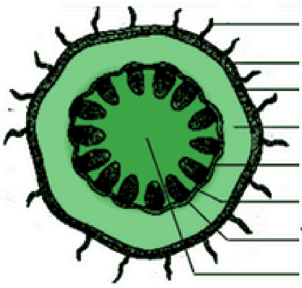
41. What is the basic unit of the nervous system? describe the function of each of its components.

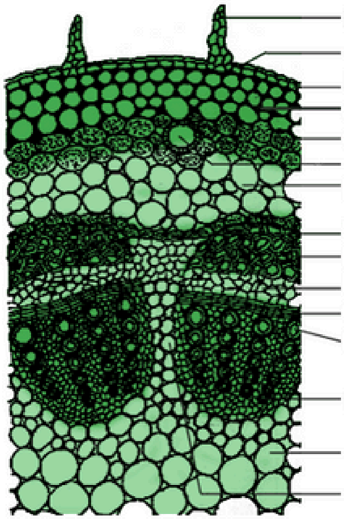
(read paragraph first)

Case Study/Source Based Question

42. Take a plant stem and cut into very thin slices or sections with the help of your teacher.
Stain the slices with safranin and place one neatly cut section on a slide and put a drop of glycerine.
Cover the slide with a cover slip and observe the cut section under the microscope. Carefully, observe the different types of cells and the arrangement of the cells
Answer the following based on your observation.

1. **Are all cells similar in structure?**
2. **How many types of cells are seen through the microscope?**
3. **Why there are so many types of cells? Can we think of reasons?**





(yaha Marks katate h)

Assertion And Reasons Questions

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

43.Assertion (A): The permeability of the cells play an important role in regulating the exchange of materials between the body
Reason (R): and the external environment. Anything entering or leaving the body must cross at least one layer of epithelium.

44. Assertion (A): Bone cells are embedded in a hard matrix.
Reason (R): The matrix found in our body may be jelly-like fluid, dense or rigid.

45. Assertion (A): Parenchyma tissue is nonliving.
Reason (R): Parenchyma cells have intercellular spaces.

46. Assertion (A): Xylem and phloem are called vascular tissues.
Reason (R): Vascular tissues conduct water, mineral salts and food materials to different plant parts.

PK Special [KBC]

47. Why is the sclerenchyma tissue present in stems and not in leaves of young plants?

48. If the bark of a tree is removed in a circular fashion, what will happen to the plant? Which tissue's function is affected?

49.How does the arrangement of muscle fibers help in voluntary and involuntary movements? Compare the structure-function relationship.

50. Describe the internal structure and function of xylem tissue.
Why is lignin deposition important in xylem vessels and tracheids?

51. Explain how the structure of neuron helps in transmission of nerve impulses.
Label the parts in a diagram. How do they differ from muscle tissues in function and structure?

52.Why do desert plants have thick cuticle and sunken stomata? Which tissues are involved in this adaptation?

SOLUTIONS

MULTIPLE CHOICE TYPE QUESTIONS

1. **A**
2. **D**
3. **B**
4. **B**
5. **C**
6. **C**
7. **C**
8. **D**
9. **C**
10. **D**
11. **A**
12. **C**
13. **A**
14. **D**
15. **D**
16. **D**
17. **D**
18. **C**
19. **C**
20. **A**
21. **A**

SHIORT ANSWER TYPE QUESTIONS

22.

Feature	**Parenchyma**	**Sclerenchyma**
Cell wall	Thin and made of cellulose	Thick and lignified
Living/Dead	Living cells	Dead cells
Intercellular space	Present	Absent
Function	Storage, photosynthesis, and support	Mechanical support and rigidity
Cell shape	Oval/round	Long and narrow
Vacuole	Large central vacuole present	Usually absent

SOLUTIONS

23. (a) → Because they are actively dividing and vacuoles would interfere with cell division.
(b)→ Because cells are tightly packed and have thick, lignified walls for mechanical strength.
(c) → Due to the presence of stone cells (a type of sclerenchyma) in the pulp.
(d)→ Due to the presence of collenchyma which provides flexibility and tensile strength.
(e) → Because the husk is made of sclerenchymatous fibres which are hard and tough.

24.Connects skin to muscles.
1. ***Fills space inside organs.***
2. ***Supports internal organs.***
3. ***Repairs tissues after injury.***

25.Blood is considered a connective tissue because it connects different body parts by transporting substances. It has a fluid matrix (plasma) with cells suspended in it and is derived from mesoderm, like other connective tissues

26.Skeletal connective tissues include cartilage and bone.
- **Functions:**
  a. **Support and shape to the body.**
  b. **Protect internal organs (e.g., skull protects brain).**
  c. **Helps in movement (bones are attachment points for muscles).**

27.

Simple Tissue	Complex Tissue
Made of one type of cells	Made of more than one type of cells
Example: Parenchyma, Collenchyma, Sclerenchyma	Example: Xylem, Phloem
Performs a common function	Performs multiple functions

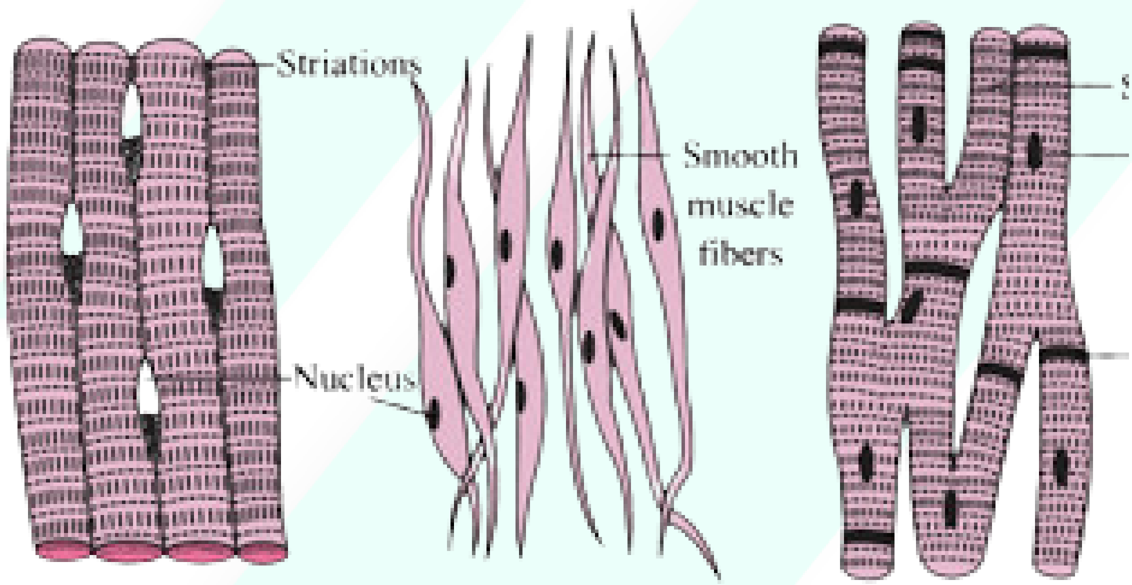
28. A plant with flexible branches has collenchyma tissue, which provides mechanical support and flexibility. Collenchyma cells are living with unevenly thickened cell walls, allowing parts like stems and branches to bend without breaking. The rough texture of the fruit is due to sclerenchyma tissue, which has dead cells with thick, lignified walls. Sclerenchyma gives hardness and protection to fruits and seeds.

SOLUTIONS

29. Structure: Made of neurons (nerve cells) and supporting cells. Neurons have a cell body, axon, and dendrites.
Function: Transmits electrical signals (impulses) between body parts and the brain.
Location: Brain, spinal cord, and nerves.

30.

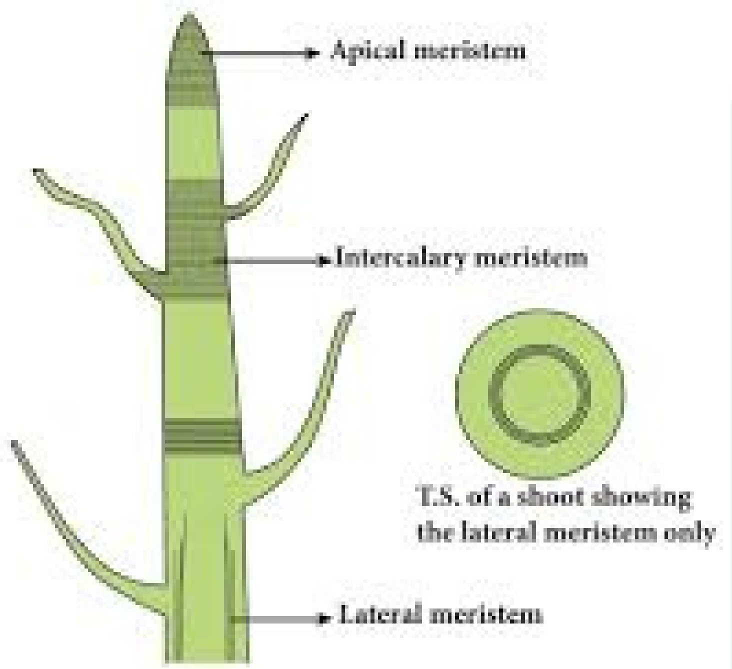
Muscle Type	Striated (Skeletal)	Smooth	Cardiac
Appearance	Striped (banded)	No striations	Striated (faint)
Control	Voluntary	Involuntary	Involuntary
Nucleus	Multinucleated	Single nucleus	One or two nuclei
Shape	Long, cylindrical	Spindle-shaped	Branched
Location	Attached to bones	Internal organs (intestine)	Heart



31. Types:

1. **Apical meristem – Tips of roots and shoots → causes length growth**
2. **Lateral meristem – Sides of stem and root → increases girth**
3. **Intercalary meristem – Base of leaves or nodes → helps in regrowth**

SOLUTIONS



32. Water hyacinth floats because it has spongy, air-filled parenchyma tissue (aerenchyma) in its petioles and leaves. This tissue helps trap air and provides buoyancy, allowing the plant to float on water even without roots in soil.

33. (A) We get a crunchy and granular feeling when we chew pear fruit:
This is because pear fruit contains sclereids, also known as stone cells, which are a type of sclerenchyma tissue. These cells have thick, lignified cell walls and are scattered throughout the soft tissue of the fruit. Their hardness and rigidity create a gritty and crunchy texture when we chew the fruit.

(B) Branches of a tree move and bend freely in high wind velocity:
Tree branches can bend and move freely in the wind due to the presence of collenchyma tissue. Collenchyma cells have unevenly thickened cell walls, which provide both mechanical support and flexibility to the plant. This allows the branches to bend without breaking during strong winds.

(C) Intercellular spaces are absent in sclerenchymatous tissues:
In sclerenchyma tissues, the cells are closely packed and have thick, lignified cell walls. These thick walls and tight packing eliminate intercellular spaces, making the tissue rigid and strong. This helps the plant in providing mechanical strength and structural support.

SOLUTIONS

34..Figure 1-T.S. collenchyma,
Figure 2-T.S. parenchyma,
Figure 3-T.S. sclerenchyma fibres.
(b) Tissue which has heavy deposition of lignin is sclerenchyma .
(c) Tissue which provides both mechanical strength and flexibility is collenchyma.
(d) Parenchyma is modified to arenchyma having air cavities in the aquatic plants.
(e) Fibres of Hemp and Jute are obtained from the sclerenchyma fibres.

35. (A)Found only in the heart – Makes up the walls of the heart.

- Involuntary in action – Works automatically without conscious control

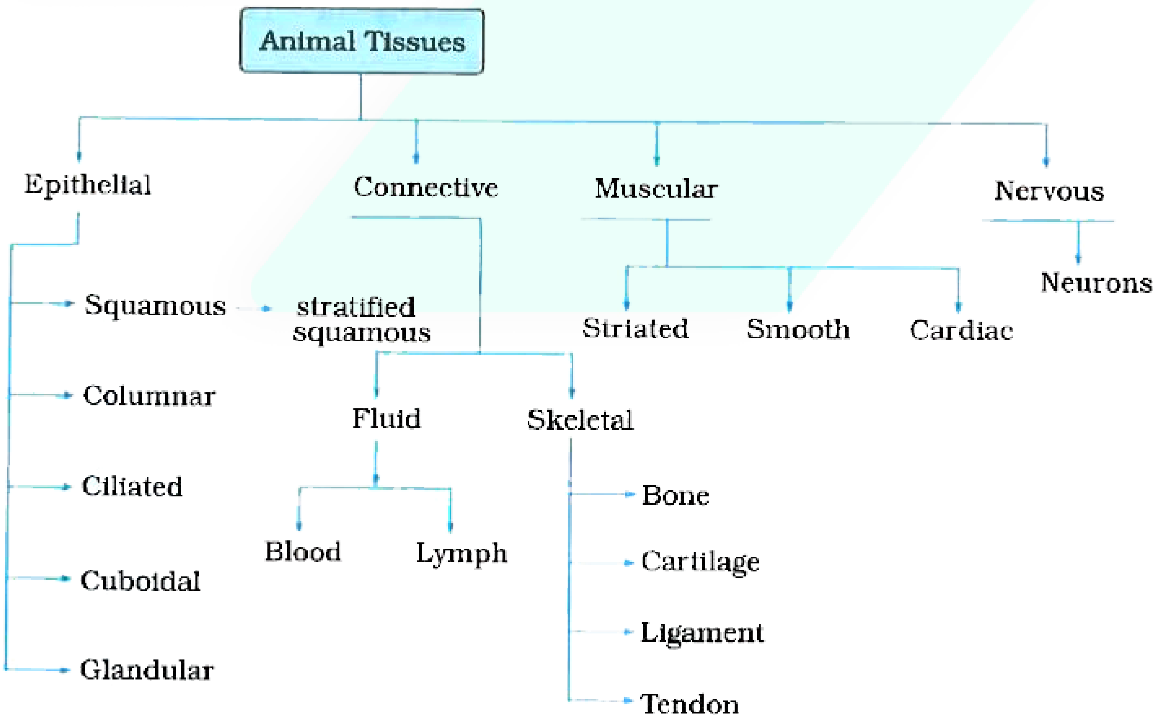
(B)Cardiac muscle tissue is found only in the walls of the heart.

(C) Cardiac muscles are highly fatigue-resistant and usually do not undergo fatigue under normal conditions.
If they do fatigue or fail, it can lead to serious heart conditions such as heart failure,

36. yes, Seema's statement is correct. The husk of a coconut is made up of sclerenchyma fibres, which are dead cells with very thick, lignified cell walls. These fibres are tough, hard, and tightly packed, which gives them great mechanical strength and makes the husk very difficult to pull out or tear. Therefore, the difficulty in removing the coconut husk is due to the presence of sclerenchymatous tissue, as correctly stated by Seema.

LONG ANSWER QUESTIONS

37.



SOLUTIONS

38. Connective tissue is a type of animal tissue that connects, binds, supports, and anchors other tissues and organs of the body. It consists of three main components: cells, fibres, and matrix (ground substance). The matrix can be solid, semi-solid, or fluid, which gives connective tissue its diverse nature.
Types of Connective Tissue:
Connective tissues are broadly classified into the following types:
1. Loose Connective Tissue
- Areolar Tissue: Found between skin and muscles, around blood vessels, and nerves.
  - Function: Connects skin to muscles, fills space inside organs, repairs tissues.
- Adipose Tissue: Found below the skin, around kidneys, and eyeballs.
  - Function: Stores fat, insulates the body, acts as energy reserve.

2. Dense Connective Tissue
- Tendons: Connect muscle to bone.
  - Function: Provide strength and flexibility (non-elastic).
- Ligaments: Connect bone to bone.
  - Function: Provide flexibility and support (elastic).

3. Skeletal Connective Tissue
- Cartilage: Found in nose, ear, trachea, and joints.
  - Function: Provides support and flexibility.
- Bone: Forms the skeleton of the body.
  - Function: Provides structural framework, protects organs, stores minerals like calcium.

4. Fluid Connective Tissue
- Blood: Composed of RBCs, WBCs, platelets, and plasma.
  - Function: Transports oxygen, nutrients, hormones, and waste products.
- Lymph: A colourless fluid that drains into the blood.
  - Function: Helps in immunity and exchange of materials between blood and tissues.

39. Permanent tissues are those tissues in plants that have lost the ability to divide. They are formed from meristematic tissues and are specialised for specific functions such as protection, support, conduction, and photosynthesis.
Types of Permanent Tissue:
Permanent tissues are classified into:
A. Simple Permanent Tissue
Made of one type of cell and performs similar functions.

SOLUTIONS

- Function: Stores food and performs photosynthesis (in chlorenchyma); gives buoyancy in aquatic plants (in aerenchyma).

1. Collenchyma
   - Structure: Living cells with unevenly thickened corners; little intercellular space.
   - Function: Provides flexibility and mechanical support in growing parts.
2. Sclerenchyma
   - Structure: Dead cells with thick lignified walls; no intercellular spaces.
   - Function: Provides strength and rigidity to the plant.

B. Complex Permanent Tissue

Made of more than one type of cell working together for a common function.

3. Xylem (Conducts water and minerals)
   - Components: Tracheids, vessels, xylem fibres, xylem parenchyma.
   - Function: Upward conduction of water and minerals from roots to shoots.
4. Phloem (Conducts food)
   - Components: Sieve tubes, companion cells, phloem fibres, phloem parenchyma.
   - Function: Transport of food from leaves to other parts of the plant.

40.(A) Epithelial tissue — specifically ciliated epithelium.

(B) Based on the typical structure of ciliated epithelial tissue:

(i) → Cilia
(ii) → Epithelial cell nucleus
(iii) → Epithelial cells
(iv) → Basement membrane

(C) Ciliated epithelial tissue consists of columnar cells with fine hair-like structures called cilia on their surface. These cilia help in moving particles or mucus in a particular direction.
In the respiratory tract, it moves mucus and trapped dust out of the lungs.
In the fallopian tubes, it helps move the ovum toward the uterus.
(D) here are four main types of animal tissues:
1. Epithelial Tissue – Covers body surfaces and lines cavities.
2. Connective Tissue – Supports and binds other tissues (e.g., bone, blood ,cartilage).
3. Muscular Tissue – Responsible for movement (skeletal, smooth, cardiac).
4. Nervous Tissue – Conducts nerve impulses and coordinates body activities.

SOLUTIONS

42. A neuron is the basic and functional unit of the nervous system. It helps in the transmission of messages as electrical impulses.
Parts of a Neuron:
Cell Body (Cyton):
Has a nucleus and cytoplasm. It controls all activities of the neuron.
Dendrites:
Short, branch-like parts that receive messages from other neurons or sensory organs.
Axon:
A long, tail-like part that carries messages away from the cell body to muscles or glands (effectors).
Function: Neurons help in quick transmission of messages in the body.

CASE STUDY/SOURCE BASED QUESTION:

42. No, all the cells are not similar in structure.
Under the microscope, we observe different shapes, sizes, and arrangements of cells. Some cells are thick-walled and compact, while others are thin-walled and loosely packed. This indicates the presence of different types of plant tissues.

Several different types of cells can be observed, including:

1. Parenchyma cells – thin-walled, living cells with large vacuoles.
2. Collenchyma cells – living cells with unevenly thickened corners.
3. Sclerenchyma cells – dead cells with thick, lignified walls (appear deep red with safranin).
4. Xylem – includes vessels and tracheids (dead, thick-walled).
5. Phloem – includes sieve tubes and companion cells (living).
6. Epidermis – outer protective single layer of cells.
7. Cortex and pith region – composed mostly of parenchyma.

Yes, the reason for the presence of so many types of cells is specialisation and division of labour in plants.

- Different cells and tissues have different functions such as:
  - Parenchyma for storage and photosynthesis.
  - Collenchyma for flexibility and support.
  - Sclerenchyma for mechanical strength.
  - Xylem for water transport.

SOLUTIONS

- Phloem for food transport.
- These specialised cells work together to help the plant grow, support its body, transport materials, and survive in different conditions.

Thus, the diversity in cell types allows the plant to function efficiently and perform multiple life activities.

ASSERTION AND REASONS TYPE QUESTIONS:

43. (a)
44. (b)
45. (d)
46. (a)

PK Special [KBC]

47. Sclerenchyma is a type of simple permanent tissue that provides mechanical strength and rigidity to plants. It is made up of dead cells with thick, lignified walls. In stems, sclerenchyma is present to support the plant and help it stand upright, especially in older and mature parts. However, young leaves are still growing and require flexibility for expansion and photosynthesis. The presence of rigid sclerenchyma would hinder their growth and flexibility. Hence, collenchyma and parenchyma are more common in young leaves, while sclerenchyma appears later as the leaf matures.

48. If the bark of a tree is removed in a circular strip (a process called girdling or ringing), it disrupts the phloem tissue located just below the bark. Phloem is responsible for the transport of food (mainly sugars) from the leaves to the rest of the plant, including the roots. Without this supply of food, the roots will die, and eventually the entire plant will die due to the failure in food transport. Thus, phloem tissue is the one whose function is affected, and it leads to the death of tissues below the ring.

SOLUTIONS

49. Voluntary muscles (also called skeletal or striated muscles) are long, cylindrical, multinucleated, and striated. These muscles are arranged in bundles and are connected to bones. Their structure allows for quick and powerful contraction, which is under conscious control. This helps in activities like walking, writing, and lifting objects.
Involuntary muscles, such as smooth muscles and cardiac muscles, are spindle-shaped (smooth) or branched (cardiac) and have one nucleus. These muscles contract slowly and rhythmically without conscious control. They are found in internal organs like the stomach, intestines (smooth), and the heart (cardiac).
Comparison:
Voluntary muscles → Rapid, controlled movement (due to striated, bundled structure)
Involuntary muscles → Slow, automatic functions (due to smooth or branched structure)
Thus, the structure of muscle fibers is directly related to their function in body movement and control.

50. Xylem is a complex permanent tissue in plants responsible for the transport of water and minerals from the roots to other parts of the plant. It also provides mechanical strength.
Xylem Components:
1. Tracheids – Long, narrow dead cells with thick lignified walls and tapering ends.
2. Vessels – Wider, tube-like dead cells joined end-to-end to form long pipelines.
3. Xylem Fibres – Dead, thick-walled supportive cells.
4. Xylem Parenchyma – Living cells that store food and help in lateral conduction.

Function:
- Conducts water and mineral salts from roots to shoots.
- Provides structural support to the plant.

Importance of Lignin Deposition:
Lignin is a complex organic substance that is deposited in the cell walls of tracheids and vessels. It makes the walls thick, rigid, and waterproof, which prevents them from collapsing under the pressure of water transport. It also helps maintain the integrity and strength of xylem tissues.

51. Neuron is the basic structural and functional unit of the nervous system. It is specialised to transmit nerve impulses quickly and efficiently across long distances.
Structure of a Neuron:
5. Dendrites – Short, branched projections that receive signals.
6. Cell Body (Cyton) – Contains the nucleus and cytoplasm; integrates signals.

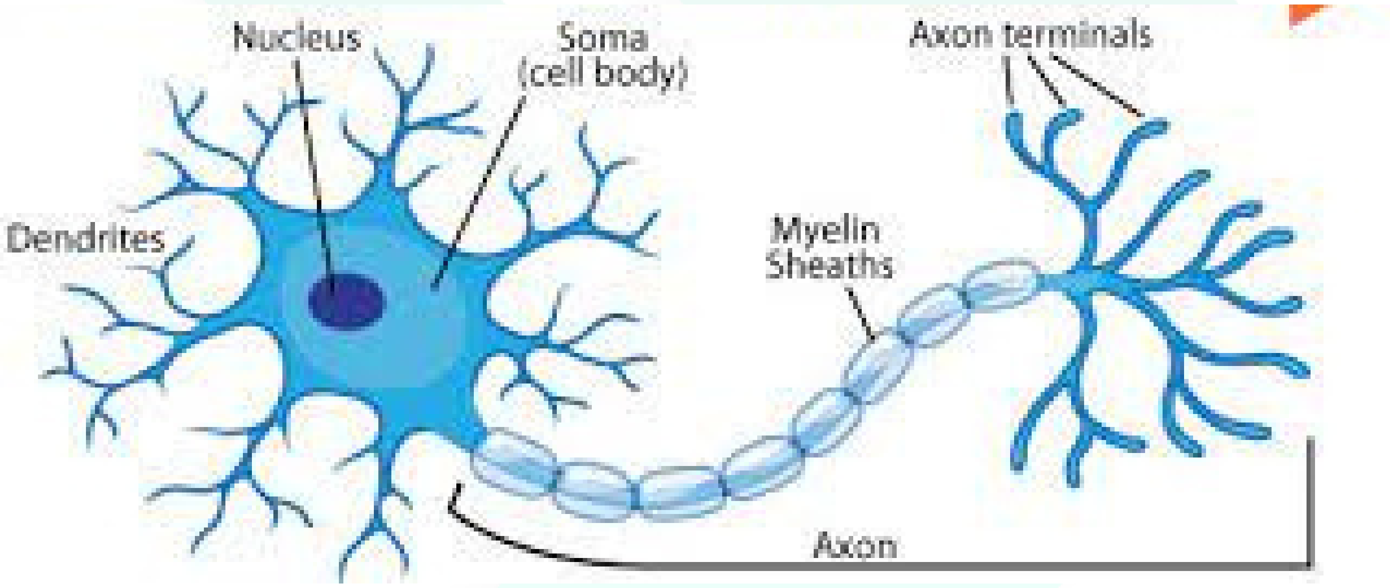
SOLUTIONS

1. Axon – A long, tubular extension that carries impulses away from the cell body.
2. Axon Terminals – Pass signals to other neurons or effectors.
3. Myelin Sheath – Fatty layer covering the axon, increases impulse speed.

Function:
Neurons receive, process, and transmit electrical impulses to other neurons, muscles, or glands.

Difference from Muscle Tissues:

- Neurons transmit electrical signals; muscle tissues contract to produce movement.
- Neurons have branched and long extensions (axon, dendrites); muscle cells are elongated and fibrous.
- Neuron signals are electrical and chemical, while muscles work via mechanical contraction.



52. Desert plants (xerophytes) are adapted to conserve water in harsh, dry environments. They have:

- Thick cuticle: A waxy layer on the epidermis that prevents excessive water loss through evaporation.
- Sunken stomata: Stomata are located in pits to trap moisture and reduce transpiration.

Tissues Involved:

- Epidermal tissue (protective tissue) forms the outer covering with cuticle.
- Parenchyma tissue in leaves stores water and may assist in photosynthesis.

These adaptations help desert plants survive in conditions where water is scarce.